Mississippi Valley State University

# CS 341 — Discrete Structures

Practice Set 47: Approved Solutions

*Solving Linear Recurrences*

Covers *MCS* Section 21.3

**Status of this set**

**Ungraded practice.** Solutions released with it. **Check your closed form against the first few terms** — it costs one line and catches almost every error.

# Part I: How to Practice This Material

**Homogeneous: three steps**

1. Write the characteristic equation and find its roots.
2. Each simple root $r$ gives a term $cr^n$; a root of multiplicity $k$ gives $r^n, nr^n, \ldots, n^{k-1}r^n$.
3. Apply the boundary conditions to solve for the constants.

**Inhomogeneous: add two steps**

Between steps 2 and 3, find a *particular* solution by guessing a form that matches $g(n)$ — a constant for a constant, a polynomial of the same degree for a polynomial, $cr^n$ for an exponential. Add it to the homogeneous solution, *then* apply the boundary conditions.

**Three mistakes to expect**

1. **Applying the boundary conditions before adding the particular solution.** They must be applied to the *general* solution, last.
2. **Using two copies of $r^n$ for a repeated root.** Use $r^n$ and $nr^n$.
3. **Sign errors reading off the characteristic equation.** $f(n) = 5f(n-1) - 6f(n-2)$ gives $x^2 - 5x + 6 = 0$.

# Part II: Practice Problems

## Section A — Distinct roots

**1.** Solve $f(n) = f(n-1) + 6f(n-2)$ with $f(0) = 1$ and $f(1) = 8$.

**Approved solution**

**Characteristic:** $x^2 = x + 6$, that is $x^2 - x - 6 = (x-3)(x+2) = 0$, with roots 3 and $-2$.
**General solution:** $f(n) = c \cdot 3^n + d \cdot (-2)^n$.
**Boundary conditions:** $c + d = 1$ and $3c - 2d = 8$. Solving gives $c = 2$, $d = -1$.

$$f(n) = 2 \cdot 3^n - (-2)^n$$

**Check** $n = 2$: formula gives $18 - 4 = 14$; the recurrence gives $8 + 6 = 14$. ✓

A negative root is perfectly acceptable, and $(-2)^n$ alternates sign. Students sometimes discard it as an error.

## Section B — Repeated roots

**2.** Solve $f(n) = 6f(n-1) - 9f(n-2)$ with $f(0) = 1$ and $f(1) = 9$.

**Approved solution**

**Characteristic:** $x^2 - 6x + 9 = (x-3)^2 = 0$, so 3 is a root of multiplicity two.
**General solution:** $f(n) = c \cdot 3^n + d \cdot n3^n$.
**Boundary conditions:** $f(0) = c = 1$; $f(1) = 3 + 3d = 9$ gives $d = 2$.

$$f(n) = 3^n + 2n \cdot 3^n$$

**Check** $n = 2$: formula gives $9 + 36 = 45$; the recurrence gives $6(9) - 9(1) = 45$. ✓

**Without the $n3^n$ term** the general solution would be just $c \cdot 3^n$, with one constant and two conditions to satisfy — impossible unless the data happen to agree.

## Section C — Inhomogeneous

**3.** Solve $f(n) = 2f(n-1) + 3$ with $f(0) = 1$.

**Approved solution**

**Step 1–2.** Homogeneous part $f(n) = 2f(n-1)$ has characteristic $x = 2$, so the homogeneous solution is $c \cdot 2^n$.
**Step 3.** $g(n) = 3$ is constant, so guess $f(n) = k$. Substituting: $k = 2k + 3$, giving $k = -3$. A particular solution is $-3$.
**Step 4.** General solution: $f(n) = c \cdot 2^n - 3$.
**Step 5.** $f(0) = c - 3 = 1$ gives $c = 4$.

$$f(n) = 4 \cdot 2^n - 3 = 2^{n+2} - 3$$

**Check:** $1, 5, 13, 29, 61$ from the recurrence; $1, 5, 13, 29, 61$ from the formula. ✓

**4.** Solve $f(n) = 3f(n-1) + 2$ with $f(0) = 0$.

**Approved solution**

Homogeneous: $c \cdot 3^n$. Particular: guess $k$, then $k = 3k + 2$ gives $k = -1$.
General: $f(n) = c \cdot 3^n - 1$. With $f(0) = c - 1 = 0$ we get $c = 1$.

$$f(n) = 3^n - 1$$

**Check:** $0, 2, 8, 26, 80$ both ways. ✓

Compare Problem 3: same shape, different constants, and the answer came out clean. The method does not care.

## Section D — Writing

**5.** In five to eight complete sentences, explain why guessing $f(n) = x^n$ is the right first move for a homogeneous linear recurrence, and what the characteristic equation is actually telling you.

**Approved solution**

**Model answer.** A homogeneous linear recurrence expresses each term as a fixed linear combination of earlier terms, and an exponential is the one kind of sequence where every term is a fixed multiple of the one before it, so exponentials are natural candidates. Substituting $f(n) = x^n$ and dividing through by the lowest power leaves a polynomial equation in $x$ alone, with the recurrence's own coefficients — the characteristic equation. Its roots are exactly the values of $x$ for which $x^n$ really does satisfy the recurrence, so each root hands you one genuine solution. By Theorem 21.3.1 any linear combination of solutions is again a solution, which turns those few roots into a whole family, and the boundary conditions then select the one member of that family matching the given data. The characteristic equation is therefore doing the real work: it converts a statement about an infinite sequence into a finite polynomial problem.

**Full credit requires** the point that the roots give solutions and Theorem 21.3.1 turns them into a family. An answer describing the procedure without explaining the guess earns about half.

**Instructor notes**

- **Problem 1's negative root** is discarded by students who expect growth. Point out that $(-2)^n$ alternates and is a perfectly good solution.
- **Problem 2's closing remark** explains why repeated roots need $nr^n$: one constant cannot meet two conditions.
- **Problems 3 and 4 are the same shape**, which lets students see the method is mechanical.
- **Insist on the check against early terms.** One line, and it catches nearly every arithmetic error.